YUVA INTERNSHIP

WEEK 2 TASK

Exploratory Data Analysis (EDA) and Visualization Framework Design

Prepared By	Yash Patil
Program	MCA
Internship	YUVA Internship
Task	Week 2 — EDA & Visualization Framework Design
Estimated Effort	33 Hours

A planning-stage framework for reliable, interpretable, and reusable EDA and visualization. No dataset-specific findings are assumed in this Week 2 task.

1. Introduction to Exploratory Data Analysis

Exploratory Data Analysis (EDA) is the systematic process of inspecting, summarizing, visualizing, and questioning a dataset before formal statistical modeling or machine learning. EDA helps a data scientist understand structure, identify quality problems, discover patterns, detect unusual observations, and formulate useful hypotheses.

Because this Week 2 task does not provide a dataset, the objective is to design a reusable EDA framework rather than report findings from a particular dataset. The framework can be adapted to education, finance, retail, logistics, customer analytics, or other domains.

Why EDA matters: a model trained on incomplete, inconsistent, biased, or poorly understood data can produce misleading results. EDA therefore acts as a quality checkpoint before modeling.

2. Objectives of the EDA Framework

• Understand dataset structure, size, schema, and variable meaning.

• Identify data types, missing values, duplicates, invalid values, and inconsistencies.

• Summarize numeric and categorical variables using suitable descriptive statistics.

• Perform univariate, bivariate, and multivariate analysis.

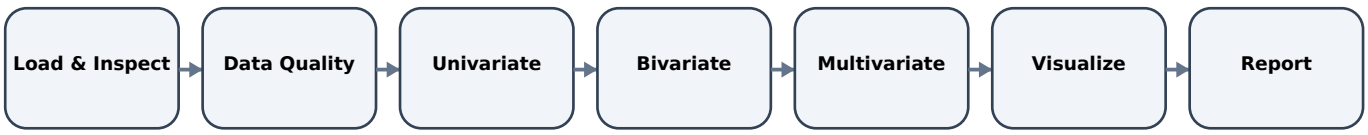
• Detect and investigate outliers without automatically deleting valid observations.

• Select visualizations that reveal distributions, comparisons, trends, and relationships.

• Document findings, assumptions, limitations, and preprocessing decisions.

• Create a reproducible framework that can be applied to future datasets.

3. EDA Workflow



Reusable end-to-end EDA lifecycle

4. Potential Data Types

Correct data-type identification determines which statistics, preprocessing methods, and charts are appropriate.

Data Type	Examples	Typical Analysis / Visualization
Numeric — Continuous	Age, salary, price, temperature	Mean/median, histogram, box plot, scatter plot
Numeric — Discrete	Count, quantity, number of orders	Frequency, bar chart, histogram
Categorical — Nominal	City, product type, department	Frequency table, bar chart
Categorical — Ordinal	Low/Medium/High, ratings	Ordered bar chart, box plot
Datetime	Date, timestamp, month	Line chart, time grouping
Boolean	Yes/No, True/False	Count plot, proportions
Text	Reviews, descriptions	Length/frequency analysis; NLP later if required

5. Initial Data Inspection

- Inspect first and last records.
- Record rows, columns, variable names, data types, and memory usage.
- Check unique values and category frequencies.
- Generate descriptive statistics for numeric and categorical columns.
- Measure missing-value counts and percentages.
- Check duplicate rows and repeated identifiers where applicable.
- Validate expected ranges, units, date formats, and business rules.
- Create a data dictionary with variable, type, unit, and analytical role.

6. Data Quality Assessment

Check	What to Examine	Planned Response
Missing data	Count and percentage by column/group	Impute, retain, flag, or remove with justification
Duplicates	Exact duplicate rows / repeated IDs	Remove or investigate based on meaning
Invalid values	Impossible ranges or categories	Correct, convert to missing, or exclude
Inconsistent formats	Dates, text case, units, spelling	Standardize formats and categories
Outliers	Extreme values and unusual combinations	Investigate before treatment
Data leakage	Information unavailable at prediction time	Exclude from predictive features
Sampling issues	Unbalanced or unrepresentative groups	Document limitations; assess sampling

7. Handling Missing Data

Missing values will first be quantified and visualized rather than immediately replaced. Treatment depends on the variable's meaning, missingness pattern, percentage missing, and analytical purpose. Options include justified row removal, mean/median imputation for suitable numeric variables, mode or an 'Unknown' category for suitable categorical variables, and forward/backward filling for appropriate ordered or time-series data. Advanced imputation may be considered when justified.

Validation: after treatment, recheck missingness and compare important distributions before and after imputation to identify artificial changes.

8. Outlier Detection & Treatment

An outlier is not automatically an error. The framework will distinguish data-entry errors from legitimate extreme cases. Detection will use box plots and IQR, z-scores where assumptions are reasonable, histograms, and scatter plots. Treatment may include source correction, capping, transformation, robust statistics/models, or retention of valid observations.

9. Univariate Analysis

Univariate analysis studies one variable at a time. Numeric variables will be summarized using count, mean, median, standard deviation, minimum, maximum, quartiles, and selected percentiles, supported by histograms, density plots, and box plots. Categorical variables will use counts and percentages, supported by bar/count plots.

10. Bivariate Analysis

Bivariate analysis examines relationships between two variables. Numeric–numeric pairs can use correlation and scatter plots; categorical–numeric pairs can use grouped summaries, box plots, or violin plots; categorical–categorical pairs can use contingency tables and grouped/stacked bar charts; datetime–numeric pairs can use time trends and rolling summaries.

Variable Pair	Question	Methods
Numeric + Numeric	Does one variable change with another?	Correlation, scatter plot, trend line
Categorical + Numeric	Do groups differ?	Box plot, violin plot, grouped summary
Categorical + Categorical	Are frequencies associated?	Contingency table, grouped bar chart
Datetime + Numeric	How does a measure change over time?	Line chart, time grouping

11. Multivariate Analysis

Multivariate analysis considers three or more variables simultaneously. Planned methods include correlation matrices and heatmaps for numeric variables, pair plots for selected features, grouped or faceted charts, and dimensionality-reduction visualizations when appropriate. Multivariate views will be limited to relevant variables so the output remains interpretable.

12. Visualization Strategy

Charts will be selected based on the analytical question, variable types, audience, and dataset size. The goal is not to create the maximum number of charts, but to create charts that reveal useful patterns clearly.

Chart	Primary Purpose	Typical Use
Histogram	Distribution and skewness	Numeric variable
Box Plot	Spread and outliers	Numeric variable or groups
Bar / Count Plot	Category comparison	Categorical variable
Scatter Plot	Relationship/correlation	Two numeric variables
Line Chart	Trend over time	Datetime + numeric
Heatmap	Correlation or matrix patterns	Many numeric variables
Violin Plot	Distribution by group	Numeric + categorical
Pair Plot	Multiple pairwise relationships	Selected numeric variables
Interactive Plot	Exploration with hover/zoom	Large or interactive reporting

13. Visualization Design Principles

- Use informative titles and descriptive axis labels.
- Show units, categories, time periods, and sample context.
- Choose suitable scales and avoid misleading truncation.
- Keep formatting consistent across related charts.
- Minimize clutter and unnecessary decoration.
- Highlight the key comparison without hiding the underlying data.
- Use readable text, legends, and accessible visual encodings.
- Always accompany important charts with a short interpretation.

14. Python Libraries & Tools

Tool	Contribution
Pandas	Loading, inspection, filtering, grouping, aggregation, missing-value analysis, and transformation.
NumPy	Numerical operations, arrays, mathematical calculations, and transformations.
Matplotlib	Flexible static plotting and detailed chart customization.
Seaborn	Statistical visualization for distributions, categories, relationships, and heatmaps.
Plotly	Interactive charts with hover, zoom, filtering, and exploratory dashboards.
Jupyter Notebook	Combines code, explanations, tables, charts, and observations in a reproducible workflow.
SciPy (optional)	Statistical functions and tests when deeper analysis is justified.

15. Proposed Notebook Structure

1. Objective and dataset description
2. Import libraries and configure notebook
3. Load data and initial inspection
4. Data dictionary and type validation
5. Data quality checks
6. Missing values and duplicates
7. Descriptive statistics
8. Univariate analysis
9. Bivariate analysis
10. Multivariate analysis
11. Outlier investigation
12. Key findings and modeling questions
13. Final EDA summary and limitations

16. Reporting & Documentation Plan

Report Section	Content
Executive Summary	Purpose, major findings, and high-level implications.
Dataset Overview	Source, size, variables, data dictionary, and target if applicable.
Data Quality	Missing values, duplicates, invalid values, outliers, and decisions.
Univariate Findings	Important distributions, frequencies, skewness, and unusual values.
Bivariate Findings	Key relationships and group comparisons supported by evidence.
Multivariate Findings	Correlations, interactions, segments, or higher-dimensional patterns.
Visualization Gallery	Selected charts with title, interpretation, and context.
Key Insights	Concise evidence-based findings linked to analytical questions.
Limitations	Coverage, quality, sampling, bias, and interpretation limitations.
Next Steps	Feature engineering, modeling, additional data, or validation questions.

17. Findings Documentation Template

Observation: What does the chart/statistic show? Evidence: Which metric or visualization supports it? Interpretation: What might it mean in context? Limitation: What cannot be concluded? Next Step: What should be investigated next?

18. Week 2 Work Plan — 33 Hours

The plan allocates 33 hours, within the required 30–35 hour range.

Activity	Planned Work	Hours
EDA concepts & objectives	Purpose, lifecycle, and success criteria	3
Data-type framework	Numeric, categorical, ordinal, datetime, boolean, text	2
Initial inspection	Schema, dimensions, summaries, uniqueness, data dictionary	3
Data quality	Missing values, duplicates, invalid values, consistency	4
Outlier strategy	Detection, investigation, treatment, validation	3
Univariate analysis	Statistics and distribution/frequency visualizations	3
Bivariate analysis	Relationships, comparisons, trends, associations	3
Multivariate analysis	Correlation, pair plots, grouped/faceted analysis	3
Visualization strategy	Chart selection, design principles, interpretation	3
Tools & reporting	Libraries, notebook structure, report framework	2
Documentation & review	Formatting, diagrams, quality check, submission	4
TOTAL		33

19. Challenges & Mitigation

Challenge	Risk	Mitigation
Large dataset	Slow or cluttered analysis	Sample/aggregate where appropriate; use interactive tools.
High-cardinality categories	Unreadable charts	Top-N categories or grouping.
Skewed distributions	Misleading summaries	Median/percentiles, transformations, robust plots.
Many missing values	Biased conclusions	Analyze missingness patterns and document treatment.
Extreme values	Distorted statistics	Investigate context; compare robust and standard summaries.
Chart overload	Audience confusion	Select decision-relevant visualizations only.
Correlation misuse	False causal conclusions	Describe association carefully; recommend validation.

20. Evaluation Criteria

• Depth and clarity of the EDA process.

• Logical progression from inspection to univariate, bivariate, and multivariate analysis.

• Appropriate selection and explanation of visualization methods.

• Clear handling strategy for missing data, duplicates, inconsistencies, and outliers.

• Correct explanation of Python libraries and tools.

• Well-organized reporting and documentation plan.

• Adaptability to different dataset types and domains.

• Professional presentation with readable tables and diagrams.

21. Expected Outcomes

• A reusable EDA checklist and workflow.

• A clear method for choosing visualizations based on data type and analytical question.

• A systematic data-quality and outlier-handling strategy.

• A reporting structure connecting evidence, insights, limitations, and next steps.

• Strong readiness for practical data handling and future modeling tasks.

22. Final EDA Checklist

☐ Dataset purpose and source documented
☐ Rows, columns, schema, and data types inspected
☐ Data dictionary prepared
☐ Missing values quantified
☐ Duplicates and invalid records checked
☐ Outliers investigated
☐ Descriptive statistics prepared
☐ Univariate analysis planned
☐ Bivariate analysis planned
☐ Multivariate analysis planned where appropriate

☐ Suitable charts selected and interpreted
☐ Findings and limitations documented
☐ Notebook/report structure is reproducible

23. Conclusion

This Week 2 framework establishes a structured approach to exploratory data analysis and visualization without assuming a specific dataset. It covers data understanding, quality assessment, univariate, bivariate and multivariate exploration, missing-data and outlier strategies, visualization selection, Python tooling, reporting, challenges, evaluation, and a 33-hour work plan. The framework is designed to be reusable when a real dataset becomes available.